The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

Fredrik Bierich
fredrik.bierich@gmail.com

October 16, 2026
Dataset License: CC BY-NC-SA 4.0

Abstract

In this dataset we bridge the abstraction gap between mathematics and physics by adopting a geometric perspective of trigonometry. We formalize this by addressing the arc length of a circular sector and defining the circular functions axiomatically as direct functional mappings from the physical parameters of a circular sector – namely the radius and the arc length – to its orthogonal projections. Finally, we demonstrate the utility of this framework by modeling simple harmonic motion and uniform circular motion through these geometric functions.

0.1 Circular Sector

0.1.1 Arc Length

$$r \in \mathbb{R}^+ \tag{1}$$

$$x \in [-r, r] \tag{2}$$

$$r \mapsto \overline{OA} \tag{3}$$

$$x \mapsto \overline{OB_0} \tag{4}$$

$$r \mapsto \overline{OC_n}, n \geq 0 \tag{5}$$

$$a_n \mapsto \overline{OB_n}, n \geq 0 \tag{6}$$

$$s_n \mapsto \overline{AC_n}, n \geq 0 \tag{7}$$

$$s \mapsto \operatorname{arc} AC_0 \tag{8}$$

$$s = \lim_{n \to \infty} 2^n s_n \tag{9a}$$

$$s_n = \sqrt{r^2 + r^2 - 2ra_n} \tag{9b}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \tag{9c}$$

$$a_0 = x, -r \leq x \leq r \tag{9d}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{r^2 + r^2 - 2ra_n} \tag{10a}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \tag{10b}$$

$$a_0 = x, -r \leq x \leq r \tag{10c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{11a}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \tag{11b}$$

$$a_0 = x, -r \leq x \leq r \tag{11c}$$

$$s = \lim_{n \to \infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{12a}$$

$$a_n = \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{12b}$$

$$a_0 = x, -r \leq x \leq r \tag{12c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{13a}$$

$$a_n = r \cdot \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{13b}$$

$$a_0 = x, -r \leq x \leq r \tag{13c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{14a}$$

$$a_n = r\sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{14b}$$

$$a_0 = x, -r \leq x \leq r \tag{14c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{15a}$$

$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}}, n \geq 1 \tag{15b}$$

$$a_0 = x, -r \leq x \leq r \tag{15c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{16a}$$

$$a_n = r\sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}}, n \geq 1 \tag{16b}$$

$$a_0 = x, -r \leq x \leq r \tag{16c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{17a}$$

$$a_n = r\sqrt{\frac{r + a_{n-1}}{2r}}, n \geq 1 \tag{17b}$$

$$a_0 = x, -r \leq x \leq r \tag{17c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{18a}$$

$$a_n = r\sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}}, n \geq 1 \tag{18b}$$

$$a_0 = x, -r \leq x \leq r \tag{18c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{19a}$$

$$a_n = r\sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}}, n \geq 1 \tag{19b}$$

$$a_0 = x, -r \leq x \leq r \tag{19c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{20a}$$

$$a_n = r\sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}}, n \geq 1 \tag{20b}$$

$$a_0 = x, -r \leq x \leq r \tag{20c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{21a}$$

$$a_n = r\sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2}, n \geq 1 \tag{21b}$$

$$a_0 = x, -r \leq x \leq r \tag{21c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{22a}$$

$$a_n = r \cdot \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \tag{22b}$$

$$a_0 = x, -r \leq x \leq r \tag{22c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{23a}$$

$$a_n = \frac{r\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \tag{23b}$$

$$a_0 = x, -r \leq x \leq r \tag{23c}$$

$$s = \lim_{n\to\infty} 2^n\sqrt{2r^2 - 2ra_n} \tag{24a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{24b}$$

$$a_0 = x, -r \leq x \leq r \tag{24c}$$

0.1.2 Area

$$r \in \mathbb{R}^+ \tag{25}$$

$$s \in \mathbb{R}^+ \tag{26}$$

$$r \mapsto \overline{OA} \tag{27}$$

$$s \mapsto \operatorname{arc} AC_0 \tag{28}$$

$$A \mapsto \operatorname{area} \triangledown OAC_0 \tag{29}$$

$$A = \frac{rs}{2} \tag{30}$$

0.1.3 Circular Functions

0.1.3.1 Arccosine

$$\begin{aligned} \arccos\colon \{(r,x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r\} &\to \mathbb{R} \\ (r,x) &\mapsto s \end{aligned} \tag{31}$$

$$\arccos(r,x) = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{32a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{32b}$$

$$a_0 = x \tag{32c}$$

0.1.3.2 Cosine

$$\begin{aligned} \cos\colon \{(r,s) \in \mathbb{R}^2 \mid r > 0\} &\to \mathbb{R} \\ (r,s) &\mapsto x \end{aligned} \tag{33}$$

$$s = \lim_{n\to\infty} 2^n \sqrt{2r^2 - 2ra_n} \tag{34a}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \tag{34b}$$

$$a_0 = \cos(r,s) \tag{34c}$$

$$s = \lim_{m\to\infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \tag{35a}$$

$$a_{m,n} = \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2}, 1 \leq n \leq m \tag{35b}$$

$$a_{m,0} = \cos(r,s), m \geq 0 \tag{35c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{36a}$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, 1 \leq n \leq m \tag{36b}$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{36c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{37a}$$

$$2a_{m,n-1} = 2 \cdot \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, 1 \leq n \leq m \tag{37b}$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{37c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{38a}$$

$$2a_{m,n-1} = \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2}, 1 \leq n \leq m \tag{38b}$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{38c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{39a}$$
$$2a_{m,n-1} = \sqrt{2r^2 + 2ra_{m,n}}, 1 \leq n \leq m \tag{39b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{39c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{40a}$$
$$(2a_{m,n-1})^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2, 1 \leq n \leq m \tag{40b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{40c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{41a}$$
$$2^2a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2, 1 \leq n \leq m \tag{41b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{41c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{42a}$$
$$4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2, 1 \leq n \leq m \tag{42b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{42c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{43a}$$
$$4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n}, 1 \leq n \leq m \tag{43b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{43c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{44a}$$
$$4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2, 1 \leq n \leq m \tag{44b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{44c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{45a}$$
$$4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n}, 1 \leq n \leq m \tag{45b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{45c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{46a}$$
$$\frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r}, 1 \leq n \leq m \tag{46b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{46c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{47a}$$
$$\frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, 1 \leq n \leq m \tag{47b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{47c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{48a}$$
$$\frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, 1 \leq n \leq m \tag{48b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{48c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{49a}$$
$$\frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r}, 1 \leq n \leq m \tag{49b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{49c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{50a}$$
$$\frac{2a_{m,n-1}^2}{r} - r = a_{m,n}, 1 \leq n \leq m \tag{50b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{50c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{51a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{51b}$$
$$2^m\sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0 \tag{51c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{52a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{52b}$$
$$\frac{2^m\sqrt{2r^2 - 2ra_{m,0}}}{2^m} = \frac{s}{2^m}, m \geq 0 \tag{52c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{53a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{53b}$$
$$\sqrt{2r^2 - 2ra_{m,0}} = \frac{s}{2^m}, m \geq 0 \tag{53c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{54a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{54b}$$
$$\left(\sqrt{2r^2 - 2ra_{m,0}}\right)^2 = \left(\frac{s}{2^m}\right)^2, m \geq 0 \tag{54c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{55a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{55b}$$
$$2r^2 - 2ra_{m,0} = \left(\frac{s}{2^m}\right)^2, m \geq 0 \tag{55c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{56a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{56b}$$
$$2r^2 - 2ra_{m,0} = \frac{s^2}{\left(2^m\right)^2}, m \geq 0 \tag{56c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{57a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{57b}$$
$$2r^2 - 2ra_{m,0} = \frac{s^2}{2^{2m}}, m \geq 0 \tag{57c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{58a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{58b}$$
$$2r^2 - 2ra_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0 \tag{58c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{59a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{59b}$$
$$-2ra_{m,0} = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0 \tag{59c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{60a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{60b}$$
$$\frac{-2ra_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}, m \geq 0 \tag{60c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{61a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \leq n \leq m \tag{61b}$$
$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}, m \geq 0 \tag{61c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{62a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{62b}$$
$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r}, m \ge 0 \tag{62c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{63a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{63b}$$
$$a_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r}, m \ge 0 \tag{63c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{64a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{64b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r}, m \ge 0 \tag{64c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{65a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{65b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r}, m \ge 0 \tag{65c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{66a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{66b}$$
$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + r, m \ge 0 \tag{66c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{67a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{67b}$$
$$a_{m,0} = r + \frac{s^2}{-2^{2m+1}r}, m \ge 0 \tag{67c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{68a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{68b}$$
$$a_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r}\right), m \ge 0 \tag{68c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{69a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{69b}$$
$$a_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r}\right), m \ge 0 \tag{69c}$$

$$\cos(r,s) = \lim_{m\to\infty} a_{m,m} \tag{70a}$$
$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{70b}$$
$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{70c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{71a}$$
$$r\hat{a}_{m,n} = \frac{2(r\hat{a}_{m,n-1})^2}{r} - r, 1 \le n \le m \tag{71b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{71c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{72a}$$
$$r\hat{a}_{m,n} = \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r, 1 \le n \le m \tag{72b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{72c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{73a}$$
$$r\hat{a}_{m,n} = 2r\hat{a}_{m,n-1}^2 - r, 1 \le n \le m \tag{73b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{73c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{74a}$$
$$\frac{r\hat{a}_{m,n}}{r} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, 1 \le n \le m \tag{74b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{74c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{75a}$$
$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, 1 \le n \le m \tag{75b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{75c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{76a}$$
$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r}, 1 \le n \le m \tag{76b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{76c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{77a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - \frac{r}{r}, 1 \le n \le m \tag{77b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{77c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{78a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{78b}$$
$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \ge 0 \tag{78c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{79a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{79b}$$
$$\frac{r\hat{a}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \ge 0 \tag{79c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{80a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{80b}$$
$$\hat{a}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \ge 0 \tag{80c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{81a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{81b}$$
$$\hat{a}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \ge 0 \tag{81c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{82a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{82b}$$
$$\hat{a}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}, m \ge 0 \tag{82c}$$

$$\cos(r,s) = \lim_{m\to\infty} r\hat{a}_{m,m} \tag{83a}$$
$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \le n \le m \tag{83b}$$
$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}, m \ge 0 \tag{83c}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, 1 \leq n \leq m \tag{84a}$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}, m \geq 0 \tag{84b}$$

$$\hat{a}_{m,n} = T_2^{\circ n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), 0 \leq n \leq m \tag{85}$$

$$\hat{a}_{m,n} = T_{2^n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), 0 \leq n \leq m \tag{86}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!}\left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - 1\right)^k, 0 \leq n \leq m \tag{87}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}\left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - 1\right)^k, 0 \leq n \leq m \tag{88}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}\left(1 - \frac{s^2}{2^{2m+1}r^2} - 1\right)^k, 0 \leq n \leq m \tag{89}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}\left(-\frac{s^2}{2^{2m+1}r^2}\right)^k, 0 \leq n \leq m \tag{90}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}\left(\frac{-s^2}{2^{2m+1}r^2}\right)^k, 0 \leq n \leq m \tag{91}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \left(\frac{-s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{92}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(2 \cdot \frac{-s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{93}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-2s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{94}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-s^2}{2^{2m}r^2}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{95}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s^2}{r^2}\right)}{2^{2m}}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{96}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{2^{2m}}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{97}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{(2^2)^m}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{98}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{-\left(\frac{s}{r}\right)^2}{4^m}\right)^k \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{99}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{\left(-\left(\frac{s}{r}\right)^2\right)^k}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{100}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k\left(\left(\frac{s}{r}\right)^2\right)^k}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{101}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \tag{102}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k\left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k}\right)}{(2k)!}, 0 \leq n \leq m \tag{103}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k(2k)!}, 0 \leq n \leq m \tag{104}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(2k)!(4^m)^k}, 0 \leq n \leq m \tag{105}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1}((2^n)^2 - j^2)}{(4^m)^k}, 0 \leq n \leq m \tag{106}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \frac{\prod_{j=0}^{k-1}((2^n)^2 - j^2)}{\prod_{j=0}^{k-1} 4^m}, 0 \leq n \leq m \tag{107}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, 0 \leq n \leq m \tag{108}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{2^{2n} - j^2}{4^m}, 0 \leq n \leq m \tag{109}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^2)^n - j^2}{4^m}, 0 \leq n \leq m \tag{110}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m}, 0 \leq n \leq m \tag{111}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{112}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{113}$$

$$\cos(r, s) = \lim_{m \to \infty} r\hat{a}_{m,m} \tag{114a}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{114b}$$

$$\cos(r, s) = \lim_{m \to \infty} r\hat{a}_{m,m} \tag{115a}$$

$$r\hat{a}_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{115b}$$

$$\cos(r, s) = \lim_{m \to \infty} a_{m,m} \tag{116a}$$

$$a_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right), 0 \leq n \leq m \tag{116b}$$

$$\cos(r, s) = \lim_{m \to \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m}\right) \tag{117}$$

$$\cos(r, s) = \lim_{m \to \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \tag{118}$$

$$\cos(r, s) = \lim_{m \to \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \tag{119}$$

$$0 \leq j \leq k - 1 < 2^m \tag{120}$$

$$0 \leq j < 2^m \tag{121}$$

$$0^2 \leq j^2 < (2^m)^2 \tag{122}$$

$$0 \leq j^2 < (2^m)^2 \tag{123}$$

$$0 \leq j^2 < 2^{2m} \tag{124}$$

$$0 \leq j^2 < (2^2)^m \tag{125}$$

$$0 \leq j^2 < 4^m \tag{126}$$

$$\frac{0}{4^m} \leq \frac{j^2}{4^m} < \frac{4^m}{4^m} \tag{127}$$

$$0 \leq \frac{j^2}{4^m} < \frac{4^m}{4^m} \tag{128}$$

$$0 \leq \frac{j^2}{4^m} < 1 \tag{129}$$

$$-0 \geq -\frac{j^2}{4^m} > -1 \tag{130}$$

$$0 \geq -\frac{j^2}{4^m} > -1 \tag{131}$$

$$0 + 1 \geq -\frac{j^2}{4^m} + 1 > -1 + 1 \tag{132}$$

$$1 \geq -\frac{j^2}{4^m} + 1 > -1 + 1 \tag{133}$$

$$1 \geq 1 - \frac{j^2}{4^m} > -1 + 1 \tag{134}$$

$$1 \geq 1 - \frac{j^2}{4^m} > 0 \tag{135}$$

$$0 < 1 - \frac{j^2}{4^m} \leq 1 \tag{136}$$

$$0 < \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \leq 1 \tag{137}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|(-1)^k \left(\frac{s}{r}\right)^{2k}\right|}{|(2k)!|} \tag{138}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|(-1)^k\right| \left|\left(\frac{s}{r}\right)^{2k}\right|}{|(2k)!|} \tag{139}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|\left(\frac{s}{r}\right)^{2k}\right|}{|(2k)!|} \tag{140}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|\frac{s}{r}\right|^{2k}}{|(2k)!|} \tag{141}$$

$$\left|\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}\right| = \frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!} \tag{142}$$

$$\lim_{k\to\infty} \frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)} = \lim_{k\to\infty} \frac{\left(\frac{\left|\frac{s}{r}\right|^{2k+2}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)} \tag{143}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2k+2}}{(2k+2)!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)} \tag{144}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=\lim_{k\to\infty}\frac{\left(\frac{(2k)!\left|\frac{s}{r}\right|^{2k+2}}{(2k+2)!}\right)}{\left|\frac{s}{r}\right|^{2k}} \tag{145}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=\lim_{k\to\infty}\frac{\left(\frac{(2k)!\left|\frac{s}{r}\right|^{2k+2}}{(2k+2)(2k+1)(2k)!}\right)}{\left|\frac{s}{r}\right|^{2k}} \tag{146}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2k+2}}{(2k+2)(2k+1)}\right)}{\left|\frac{s}{r}\right|^{2k}} \tag{147}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=\lim_{k\to\infty}\frac{\left|\frac{s}{r}\right|^{2}}{(2k+2)(2k+1)} \tag{148}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}=0 \tag{149}$$

$$\lim_{k\to\infty}\frac{\left(\frac{\left|\frac{s}{r}\right|^{2(k+1)}}{(2(k+1))!}\right)}{\left(\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}\right)}<1 \tag{150}$$

$$\sum_{k=0}^{\infty}\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!}<\infty \tag{151}$$

$$\left|\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^m}\right)\right|\leq\frac{\left|\frac{s}{r}\right|^{2k}}{(2k)!} \tag{152}$$

$$\cos(r,s)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\lim_{m\to\infty}\prod_{j=0}^{k-1}\left(1-\frac{j^2}{4^m}\right) \tag{153}$$

$$\cos(r,s)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}\lim_{m\to\infty}\left(1-\frac{j^2}{4^m}\right) \tag{154}$$

$$\cos(r,s)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}\left(1-\lim_{m\to\infty}\frac{j^2}{4^m}\right) \tag{155}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}(1-0) \tag{156}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\prod_{j=0}^{k-1}1 \tag{157}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\cdot 1^k \tag{158}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\cdot 1 \tag{159}$$

$$\cos(r,s) = r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{160}$$

0.1.3.2.1 Partial Derivative

$$\frac{\partial\cos(r,s)}{\partial s} = \frac{\partial\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k}}{(2k)!}\right)}{\partial s} \tag{161}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \frac{\partial\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\frac{s^{2k}}{r^{2k}}}{(2k)!}\right)}{\partial s} \tag{162}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \frac{\partial\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k s^{2k}}{r^{2k}(2k)!}\right)}{\partial s} \tag{163}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \frac{\partial\left(\sum\limits_{k=0}^{\infty}\frac{(-1)^k r s^{2k}}{r^{2k}(2k)!}\right)}{\partial s} \tag{164}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \frac{\partial\left(\sum\limits_{k=0}^{\infty}\frac{(-1)^k s^{2k}}{r^{2k-1}(2k)!}\right)}{\partial s} \tag{165}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \sum_{k=0}^{\infty}\frac{(-1)^k\frac{\partial s^{2k}}{\partial s}}{r^{2k-1}(2k)!} \tag{166}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \sum_{k=1}^{\infty}\frac{(-1)^k(2k)s^{2k-1}}{r^{2k-1}(2k)!} \tag{167}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \sum_{k=1}^{\infty}\frac{(-1)^k(2k)\frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \tag{168}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \sum_{k=1}^{\infty}\frac{(-1)^k(2k)\left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \tag{169}$$

$$\frac{\partial\cos(r,s)}{\partial s} = \sum_{k=1}^{\infty}\frac{(-1)^k(2k)\left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \tag{170}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{171}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{172}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \tag{173}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \tag{174}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \tag{175}$$

$$\frac{\partial \cos(r,s)}{\partial s} = -\sum_{k=0}^{\infty} \frac{(-1)^{k} \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{176}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{177}$$

$$\frac{\partial \cos(r,s)}{\partial s} = \frac{-\sin(r,s)}{r} \tag{178}$$

0.1.3.3 Arcsine

$$\begin{aligned} \arcsin\colon \{(r,y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\to \mathbb{R} \\ (r,y) &\mapsto s \end{aligned} \tag{179}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y) + \arcsin(r,y)}{2} \tag{180}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y) + \arcsin(r,y) + \arccos(r,y) - \arccos(r,y)}{2} \tag{181}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y) + \arcsin(r,-y) + \arccos(r,-y) - \arccos(r,y)}{2} \tag{182}$$

$$\arcsin(r,y) = \frac{\arcsin(r,y) - \arcsin(r,y) + \arccos(r,-y) - \arccos(r,y)}{2} \tag{183}$$

$$\arcsin(r,y) = \frac{\arccos(r,-y) - \arccos(r,y)}{2} \tag{184}$$

0.1.3.4 Sine

$$\begin{aligned} \sin\colon \{(r,s) \in \mathbb{R}^2 \mid r > 0\} &\to \mathbb{R} \\ (r,s) &\mapsto y \end{aligned} \tag{185}$$

$$\sin^2(r,\alpha-\beta) + (r - \cos(r,\alpha-\beta))^2 = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{186}$$

$$\cos^2(r,\alpha-\beta) + \sin^2(r,\alpha-\beta) + (r - \cos(r,\alpha-\beta))^2 - \cos^2(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{187}$$

$$r^2 + (r - \cos(r,\alpha-\beta))^2 - \cos^2(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{188}$$

$$r^2 + r^2 - 2r\cos(r,\alpha-\beta) + \cos^2(r,\alpha-\beta) - \cos^2(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{189}$$

$$2r^2 - 2r\cos(r,\alpha-\beta) + \cos^2(r,\alpha-\beta) - \cos^2(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{190}$$

$$2r^2 - 2r\cos(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 \tag{191}$$

$$2r^2 - 2r\cos(r,\alpha-\beta) - 2r^2 = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 - 2r^2 \tag{192}$$

$$-2r\cos(r,\alpha-\beta) = (\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 - 2r^2 \tag{193}$$

$$\frac{-2r\cos(r,\alpha-\beta)}{-2r} = \frac{(\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 - 2r^2}{-2r} \tag{194}$$

$$\cos(r,\alpha-\beta) = \frac{(\cos(r,\alpha) - \cos(r,\beta))^2 + (\sin(r,\alpha) - \sin(r,\beta))^2 - 2r^2}{-2r} \tag{195}$$

$$\cos(r,\alpha-\beta) = \frac{\cos^2(r,\alpha) - 2\cos(r,\alpha)\cos(r,\beta) + \cos^2(r,\beta) + (\sin(r,\alpha) - \sin(r,\beta))^2 - 2r^2}{-2r} \tag{196}$$

$$\cos(r,\alpha-\beta) = \frac{\cos^2(r,\alpha) - 2\cos(r,\alpha)\cos(r,\beta) + \cos^2(r,\beta) + \sin^2(r,\alpha) - 2\sin(r,\alpha)\sin(r,\beta) + \sin^2(r,\beta) - 2r^2}{-2r} \tag{197}$$

$$\cos(r,\alpha-\beta) = \frac{(\cos^2(r,\alpha) + \sin^2(r,\alpha)) - 2\cos(r,\alpha)\cos(r,\beta) + \cos^2(r,\beta) - 2\sin(r,\alpha)\sin(r,\beta) + \sin^2(r,\beta) - 2r^2}{-2r} \tag{198}$$

$$\cos(r,\alpha-\beta) = \frac{(\cos^2(r,\alpha) + \sin^2(r,\alpha)) - 2\cos(r,\alpha)\cos(r,\beta) + (\cos^2(r,\beta) + \sin^2(r,\beta)) - 2\sin(r,\alpha)\sin(r,\beta) - 2r^2}{-2r} \tag{199}$$

$$\cos(r,\alpha-\beta) = \frac{r^2 - 2\cos(r,\alpha)\cos(r,\beta) + (\cos^2(r,\beta) + \sin^2(r,\beta)) - 2\sin(r,\alpha)\sin(r,\beta) - 2r^2}{-2r} \tag{200}$$

$$\cos(r,\alpha-\beta)=\frac{r^2-2\cos(r,\alpha)\cos(r,\beta)+r^2-2\sin(r,\alpha)\sin(r,\beta)-2r^2}{-2r} \tag{201}$$

$$\cos(r,\alpha-\beta)=\frac{2r^2-2\cos(r,\alpha)\cos(r,\beta)-2\sin(r,\alpha)\sin(r,\beta)-2r^2}{-2r} \tag{202}$$

$$\cos(r,\alpha-\beta)=\frac{-2\cos(r,\alpha)\cos(r,\beta)-2\sin(r,\alpha)\sin(r,\beta)}{-2r} \tag{203}$$

$$\cos(r,\alpha-\beta)=\frac{-2\left(\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)\right)}{-2r} \tag{204}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)}{r} \tag{205}$$

$$\cos(r,\alpha-\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha-\beta}{r}\right)^{2k}}{(2k)!} \tag{206}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha-\beta}{r}\right)^{2k}}{(2k)!} \tag{207}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}-\frac{\beta}{r}\right)^{2k}}{(2k)!} \tag{208}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}+\frac{-\beta}{r}\right)^{2k}}{(2k)!} \tag{209}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\sum\limits_{j=0}^{2k}\binom{2k}{j}\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{(2k)!} \tag{210}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\sum\limits_{j=0}^{2k}\frac{(2k)!}{j!(2k-j)!}\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{(2k)!} \tag{211}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k\sum\limits_{j=0}^{2k}(2k)!\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{212}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}\frac{(-1)^k(2k)!\sum\limits_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!}}{(2k)!} \tag{213}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{j!(2k-j)!} \tag{214}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}\left(\frac{-\beta}{r}\right)^{j}}{(2k-j)!j!} \tag{215}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\sum_{k=0}^{n}(-1)^k\sum_{j=0}^{2k}\frac{\left(\frac{\alpha}{r}\right)^{2k-j}}{(2k-j)!}\frac{\left(\frac{-\beta}{r}\right)^{j}}{j!} \tag{216}$$

$$\cos(r,\alpha-\beta)=r\lim_{n\to\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{j=0}^{n-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j}}{(2j)!}+\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right) \tag{217}$$

$$\cos(r,\alpha-\beta)=r\left(\lim_{n\to\infty}\left(\sum_{k=0}^{n}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{j=0}^{n-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j}}{(2j)!}\right)+\lim_{n\to\infty}\left(\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right)\right) \tag{218}$$

$$\cos(r,\alpha-\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+\lim_{n\to\infty}\left(\sum_{k=0}^{n-1}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{j=0}^{n-1-k}\frac{(-1)^j\left(\frac{\beta}{r}\right)^{2j+1}}{(2j+1)!}\right)\right) \tag{219}$$

$$\cos(r,\alpha-\beta)=r\left(\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{220}$$

$$\cos(r,\alpha-\beta)=r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!} \tag{221}$$

$$\cos(r,\alpha-\beta)=\frac{r\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{222}$$

$$\cos(r,\alpha-\beta)=\frac{r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}+r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{223}$$

$$\cos(r,\alpha-\beta)=\frac{\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+r^2\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}}{r} \tag{224}$$

$$\cos(r,\alpha-\beta)=\frac{\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k}}{(2k)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{225}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k}}{(2k)!}\right)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{226}$$

$$\cos(r,\alpha-\beta)=\frac{\cos(r,\alpha)\cos(r,\beta)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{227}$$

$$\frac{\cos(r,\alpha)\cos(r,\beta)+\sin(r,\alpha)\sin(r,\beta)}{r}=\frac{\cos(r,\alpha)\cos(r,\beta)+\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum\limits_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right)}{r} \tag{228}$$

$$\sin(r,\alpha)\sin(r,\beta)=\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\alpha}{r}\right)^{2k+1}}{(2k+1)!}\right)\left(r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{\beta}{r}\right)^{2k+1}}{(2k+1)!}\right) \tag{229}$$

$$\sin(r,s)\overset{\{\alpha,\beta\}\mapsto s}{=}r\sum_{k=0}^{\infty}\frac{(-1)^k\left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \tag{230}$$

0.1.3.4.1 Partial Derivative

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \tag{231}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \tag{232}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(r \sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{233}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1}(2k+1)!} \right)}{\partial s} \tag{234}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\partial \left(\sum\limits_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k}(2k+1)!} \right)}{\partial s} \tag{235}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k}(2k+1)!} \tag{236}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k}(2k+1)!} \tag{237}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \tag{238}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \tag{239}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \tag{240}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \tag{241}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{r \sum\limits_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \tag{242}$$

$$\frac{\partial \sin(r,s)}{\partial s} = \frac{\cos(r,s)}{r} \tag{243}$$

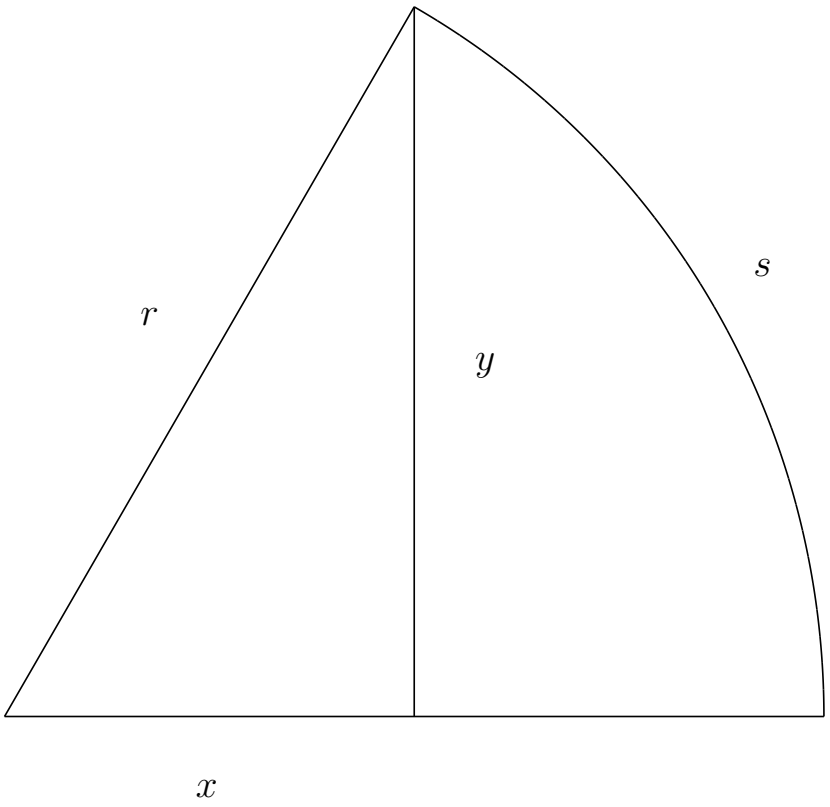


Figure 1: Circular sector.

0.2 Circular Sector Sequence

0.2.1 Apothem

$$\text{area}\triangle OAC_{n-1}=\frac{\overline{AC_{n-1}OB_n}}{2},n\geq 1 \tag{244}$$

$$\text{area}\triangle OAC_{n-1}=\frac{\overline{OC_{n-1}AB_{n-1}}}{2},n\geq 1 \tag{245}$$

$$\frac{\overline{AC_{n-1}OB_n}}{2}=\frac{\overline{OC_{n-1}AB_{n-1}}}{2},n\geq 1 \tag{246}$$

$$\overline{AC_{n-1}OB_n}=\overline{OC_{n-1}AB_{n-1}},n\geq 1 \tag{247}$$

$$\frac{\overline{AC_{n-1}OB_n}}{\overline{AC_{n-1}}}=\frac{\overline{OC_{n-1}AB_{n-1}}}{\overline{AC_{n-1}}},n\geq 1 \tag{248}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}AB_{n-1}}}{\overline{AC_{n-1}}},n\geq 1 \tag{249}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}}\sqrt{\overline{OA}^2-\overline{OB_{n-1}}^2}}{\overline{AC_{n-1}}},n\geq 1 \tag{250}$$

$$\overline{OB_n}=\frac{\overline{OC_{n-1}}\sqrt{\overline{OA}^2-\overline{OB_{n-1}}^2}}{\sqrt{\overline{OA}^2+\overline{OC_{n-1}}^2-2\overline{OC_{n-1}OB_{n-1}}}},n\geq 1 \tag{251}$$

0.2.2 Chord Length

$$\overline{AC_n}=\sqrt{\overline{AB_n}^2+\overline{B_nC_n}^2} \tag{252}$$

$$\overline{AC_n}=\sqrt{\left(\sqrt{\overline{OA}^2-\overline{OB_n}^2}\right)^2+\overline{B_nC_n}^2} \tag{253}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\overline{B_nC_n}^2} \tag{254}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\left(\overline{OC_n}-\overline{OB_n}\right)^2} \tag{255}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2-\overline{OB_n}^2+\overline{OC_n}^2-2\overline{OC_nOB_n}+\overline{OB_n}^2} \tag{256}$$

$$\overline{AC_n}=\sqrt{\overline{OA}^2+\overline{OC_n}^2-2\overline{OC_nOB_n}} \tag{257}$$

0.2.3 Arc Length

$$\text{arc}\,AC_0\overset{\text{def}}{=}\lim_{n\to\infty}2^n\overline{AC_n} \tag{258}$$

0.2.4 Area

$$\operatorname{area} \triangledown OAC_0 = \lim_{n\to\infty} 2^{n-1} \operatorname{area} OAC_nC_{n-1} \tag{259}$$

$$\operatorname{area} \triangledown OAC_0 = \lim_{n\to\infty} 2^{n-1} \frac{\overline{AC_{n-1}OC_n}}{2} \tag{260}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} 2^{n-1}\overline{AC_{n-1}OC_n}}{2} \tag{261}$$

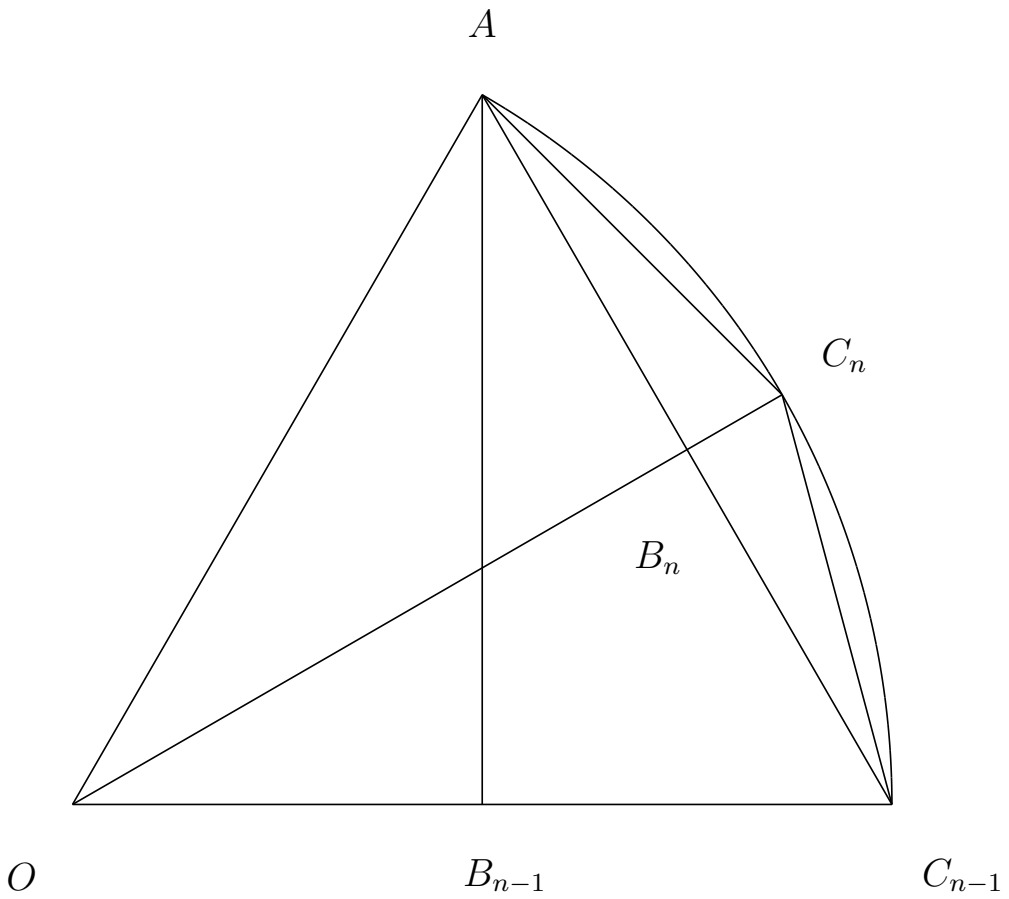
$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} \left(2^{n-1}\overline{AC_{n-1}}\right)\overline{OC_n}}{2} \tag{262}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\lim_{n\to\infty} \overline{OC_n}\left(2^{n-1}\overline{AC_{n-1}}\right)}{2} \tag{263}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\left(\lim_{n\to\infty} \overline{OC_n}\right)\left(\lim_{n\to\infty} 2^{n-1}\overline{AC_{n-1}}\right)}{2} \tag{264}$$

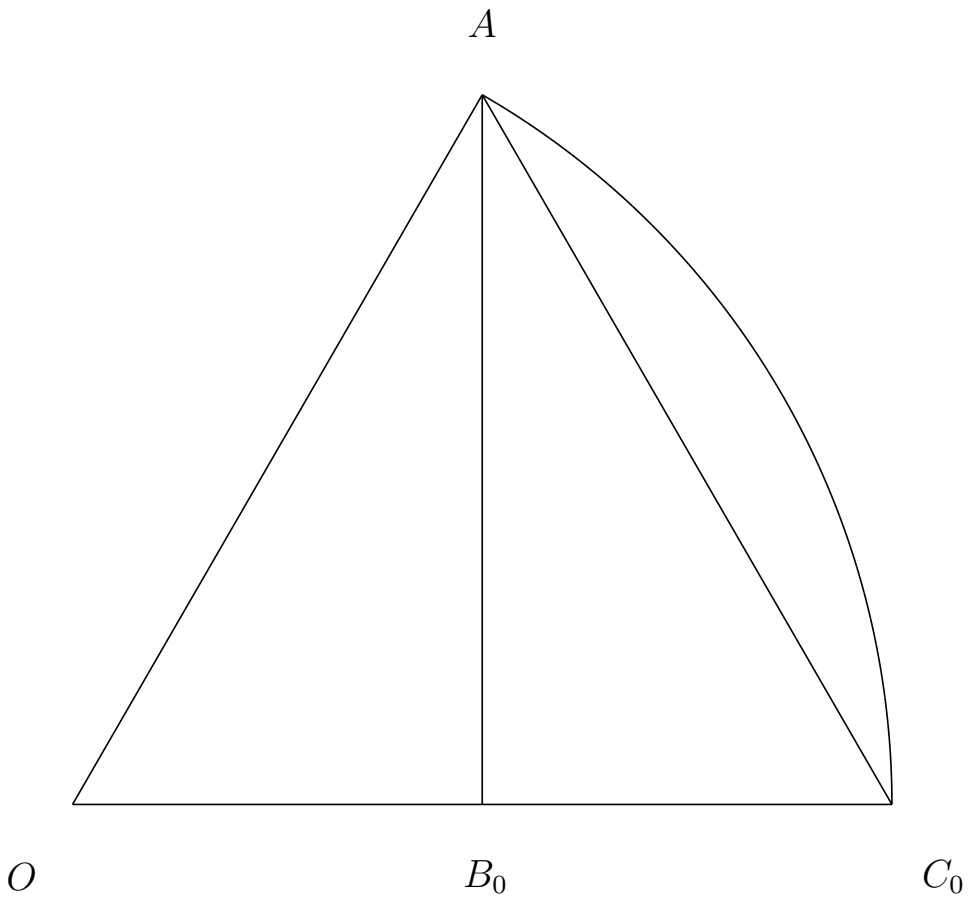
$$\operatorname{area} \triangledown OAC_0 = \frac{\overline{OA}\lim_{n\to\infty} 2^{n-1}\overline{AC_{n-1}}}{2} \tag{265}$$

$$\operatorname{area} \triangledown OAC_0 = \frac{\overline{OA}\operatorname{arc} AC_0}{2} \tag{266}$$



(a) Recursive step.

Figure 2: Circular sector sequence.



(b) Base case.

Figure 2: Circular sector sequence.

0.3 Simple Harmonic Motion

$$[m] = \text{kg} \tag{267}$$

$$[\omega] = \text{m/s} \tag{268}$$

$$[k] = \text{N/m} \tag{269}$$

$$[r] = \text{m} \tag{270}$$

$$s = \omega t \tag{271}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta s|}{|\Delta t|} \tag{272}$$

$$(\Delta s)^2 \approx (\Delta x)^2 + (\Delta y)^2 \tag{273}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx \frac{(\Delta x)^2}{(\Delta x)^2} + \frac{(\Delta y)^2}{(\Delta x)^2} \tag{274}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx 1 + \frac{(\Delta y)^2}{(\Delta x)^2} \tag{275}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} \approx 1 + \left(\frac{\Delta y}{\Delta x}\right)^2 \tag{276}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} = 1 + 0 \iff x \to 0 \tag{277}$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} = 1 \iff x \to 0 \tag{278}$$

$$(\Delta x)^2 \frac{(\Delta s)^2}{(\Delta x)^2} = (\Delta x)^2 \iff x \to 0 \tag{279}$$

$$\frac{(\Delta x)^2 (\Delta s)^2}{(\Delta x)^2} = (\Delta x)^2 \iff x \to 0 \tag{280}$$

$$(\Delta s)^2 = (\Delta x)^2 \iff x \to 0 \tag{281}$$

$$|\Delta s| = |\Delta x| \iff x \to 0 \tag{282}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta x|}{|\Delta t|} \iff x \to 0 \tag{283}$$

$$F = ma \tag{284}$$

$$F = -kx \tag{285}$$

$$ma = -kx \tag{286}$$

$$x = \cos(r, s) \tag{287}$$

$$x(t) = \cos(r, \omega t) \tag{288}$$

$$v(t) = \frac{dx}{dt} \tag{289}$$

$$v(t) = \left.\frac{\partial \cos(r, s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{290}$$

$$v(t) = \frac{-\sin(r, \omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{291}$$

$$v(t) = \frac{-\sin(r, \omega t)}{r} \cdot \omega \tag{292}$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r} \tag{293}$$

$$a(t) = \frac{dv}{dt} \tag{294}$$

$$a(t) = \frac{d\left(\frac{\partial \cos(r,s)}{\partial s}\Big|_{s=\omega t} \frac{d(\omega t)}{dt}\right)}{dt} \tag{295}$$

$$a(t) = \left(\frac{\partial^2 \cos(r, s)}{\partial s^2} \cdot \frac{ds}{dt}\Big|_{s=\omega t}\right) \cdot \frac{d(\omega t)}{dt} + \frac{\partial \cos(r, s)}{\partial s}\Big|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{296}$$

$$a(t) = \left(\frac{-\cos(r, \omega t)}{r^2} \cdot \frac{d(\omega t)}{dt}\right) \cdot \frac{d(\omega t)}{dt} + \frac{\partial \cos(r, s)}{\partial s}\Big|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{297}$$

$$a(t) = \frac{-\cos(r, \omega t)}{r^2} \cdot \left(\frac{d(\omega t)}{dt}\right)^2 + \frac{\partial \cos(r, s)}{\partial s}\Big|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{298}$$

$$a(t) = \frac{-\cos(r, \omega t)}{r^2} \cdot \omega^2 + \frac{\partial \cos(r, s)}{\partial s}\Big|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{299}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} + \frac{\partial \cos(r, s)}{\partial s}\Big|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{300}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} + \frac{-\sin(r, \omega t)}{r} \cdot \frac{d^2(\omega t)}{dt^2} \tag{301}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} + \frac{-\sin(r, \omega t)}{r} \cdot 0 \tag{302}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} + 0 \tag{303}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2} \tag{304}$$

$$a(x) = \frac{-\omega^2 x}{r^2} \tag{305}$$

$$m\frac{-\omega^2 x}{r^2} = -kx \tag{306}$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx \tag{307}$$

$$\frac{-m\omega^2 x}{r^2} = -kx \tag{308}$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2} \tag{309}$$

$$m\omega^2 = kr^2 \tag{310}$$

$$K(t) = \frac{mv^2(t)}{2} \tag{311}$$

$$K(t) = \frac{m\left(\frac{-\omega \sin(r,\omega t)}{r}\right)^2}{2} \tag{312}$$

$$K(t) = \frac{m\frac{(-\omega \sin(r,\omega t))^2}{r^2}}{2} \tag{313}$$

$$K(t) = \frac{m\frac{(-\omega)^2 \sin^2(r,\omega t)}{r^2}}{2} \tag{314}$$

$$K(t) = \frac{m\frac{\omega^2 \sin^2(r,\omega t)}{r^2}}{2} \tag{315}$$

$$K(t) = \frac{\left(\frac{m\omega^2 \sin^2(r,\omega t)}{r^2}\right)}{2} \tag{316}$$

$$K(t) = \frac{\left(\frac{kr^2 \sin^2(r,\omega t)}{r^2}\right)}{2} \tag{317}$$

$$K(t) = \frac{k \sin^2(r,\omega t)}{2} \tag{318}$$

$$U(t) = \frac{kx^2(t)}{2} \tag{319}$$

$$U(t) = \frac{k \cos^2(r,\omega t)}{2} \tag{320}$$

$$E = K + U \tag{321}$$

$$E = \frac{k \sin^2(r,\omega t)}{2} + \frac{k \cos^2(r,\omega t)}{2} \tag{322}$$

$$E = \frac{k \sin^2(r,\omega t) + k \cos^2(r,\omega t)}{2} \tag{323}$$

$$E = \frac{k(\sin^2(r,\omega t) + \cos^2(r,\omega t))}{2} \tag{324}$$

$$E = \frac{kr^2}{2} \tag{325}$$

$$E = \frac{m\omega^2}{2} \tag{326}$$

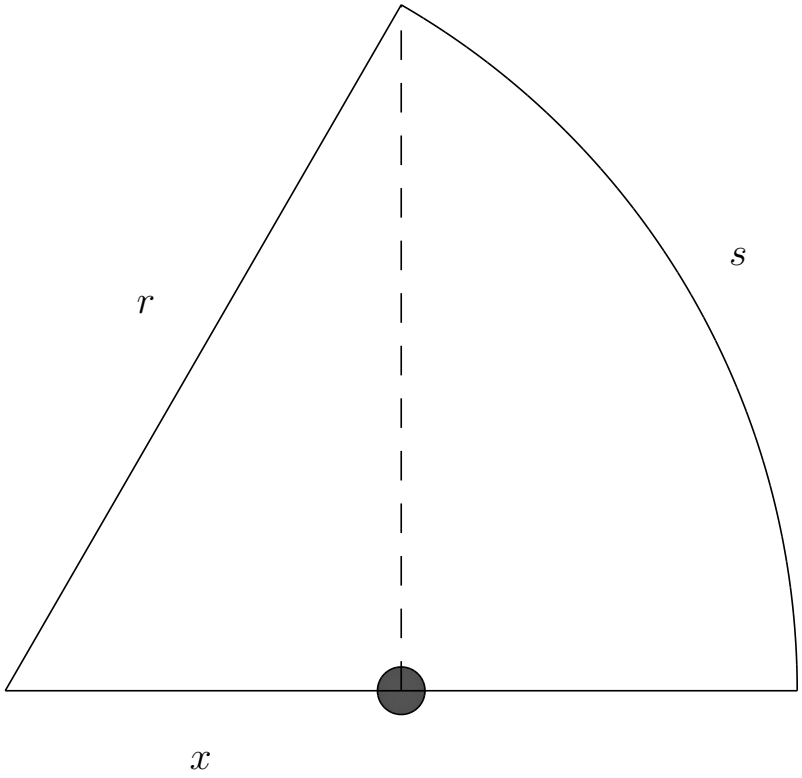


Figure 3: Simple harmonic motion.

0.4 Uniform Circular Motion

$$[m] = \text{kg} \tag{327}$$

$$[\omega] = \text{m/s} \tag{328}$$

$$[r] = \text{m} \tag{329}$$

$$s = \omega t \tag{330}$$

$$\omega = \lim_{|\Delta t| \to 0} \frac{|\Delta s|}{|\Delta t|} \tag{331}$$

$$x = \cos(r, s) \tag{332}$$

$$x(t) = \cos(r, \omega t) \tag{333}$$

$$y = \sin(r, s) \tag{334}$$

$$y(t) = \sin(r, \omega t) \tag{335}$$

$$\vec{r}(t) = (x, y) \tag{336}$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t)) \tag{337}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \tag{338}$$

$$\vec{v}(t) = \frac{d(x, y)}{dt} \tag{339}$$

$$\vec{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt} \right) \tag{340}$$

$$\frac{dx}{dt} = \left. \frac{\partial \cos(r, s)}{\partial s} \right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{341}$$

$$\frac{dx}{dt} = \frac{-\sin(r, \omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{342}$$

$$\frac{dx}{dt} = \frac{-\sin(r, \omega t)}{r} \cdot \omega \tag{343}$$

$$\frac{dx}{dt} = \frac{-\omega \sin(r, \omega t)}{r} \tag{344}$$

$$\frac{dy}{dt} = \left. \frac{\partial \sin(r, s)}{\partial s} \right|_{s=\omega t} \cdot \frac{d(\omega t)}{dt} \tag{345}$$

$$\frac{dy}{dt} = \frac{\cos(r, \omega t)}{r} \cdot \frac{d(\omega t)}{dt} \tag{346}$$

$$\frac{dy}{dt} = \frac{\cos(r, \omega t)}{r} \cdot \omega \tag{347}$$

$$\frac{dy}{dt} = \frac{\omega \cos(r, \omega t)}{r} \tag{348}$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right) \tag{349}$$

$$\vec{v}(t) = \frac{\omega \left(-\sin(r, \omega t), \cos(r, \omega t) \right)}{r} \tag{350}$$

$$v = |\vec{v}| \tag{351}$$

$$v = \left| \frac{\omega \left(-\sin(r, \omega t), \cos(r, \omega t) \right)}{r} \right| \tag{352}$$

$$v = \frac{|\omega \left(-\sin(r, \omega t), \cos(r, \omega t) \right)|}{|r|} \tag{353}$$

$$v = \frac{|\omega| \, |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \tag{354}$$

$$v = \frac{\omega \, |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \tag{355}$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|} \tag{356}$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|} \tag{357}$$

$$v = \frac{\omega \sqrt{r^2}}{|r|} \tag{358}$$

$$v = \frac{\omega \, |r|}{|r|} \tag{359}$$

$$v = \omega \tag{360}$$

$$E = \frac{mv^2}{2} \tag{361}$$

$$E = \frac{m\omega^2}{2} \tag{362}$$

$$\vec{a}(t) = \frac{d^2 \vec{r}}{dt^2} \tag{363}$$

$$\vec{a}(t) = \frac{d^2 (x, y)}{dt^2} \tag{364}$$

$$\vec{a}(t) = \left(\frac{d^2 x}{dt^2}, \frac{d^2 y}{dt^2} \right) \tag{365}$$

$$\frac{d^2 x}{dt^2} = \frac{d \left(\frac{\partial \cos(r,s)}{\partial s} \Big|_{s=\omega t} \frac{d(\omega t)}{dt} \right)}{dt} \tag{366}$$

$$\frac{d^2 x}{dt^2} = \left(\frac{\partial^2 \cos(r, s)}{\partial s^2} \cdot \frac{ds}{dt} \Big|_{s=\omega t} \right) \cdot \frac{d(\omega t)}{dt} + \frac{\partial \cos(r, s)}{\partial s} \Big|_{s=\omega t} \cdot \frac{d^2 (\omega t)}{dt^2} \tag{367}$$

$$\frac{d^2 x}{dt^2} = \left(\frac{-\cos(r, \omega t)}{r^2} \cdot \frac{d(\omega t)}{dt} \right) \cdot \frac{d(\omega t)}{dt} + \frac{\partial \cos(r, s)}{\partial s} \Big|_{s=\omega t} \cdot \frac{d^2 (\omega t)}{dt^2} \tag{368}$$

$$\frac{d^2 x}{dt^2} = \frac{-\cos(r, \omega t)}{r^2} \cdot \left(\frac{d(\omega t)}{dt} \right)^2 + \frac{\partial \cos(r, s)}{\partial s} \Big|_{s=\omega t} \cdot \frac{d^2 (\omega t)}{dt^2} \tag{369}$$

$$\frac{d^2 x}{dt^2} = \frac{-\cos(r, \omega t)}{r^2} \cdot \omega^2 + \frac{\partial \cos(r, s)}{\partial s} \Big|_{s=\omega t} \cdot \frac{d^2 (\omega t)}{dt^2} \tag{370}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \left.\frac{\partial\cos(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{371}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r} \cdot \frac{d^2(\omega t)}{dt^2} \tag{372}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + \frac{-\sin(r,\omega t)}{r} \cdot 0 \tag{373}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} + 0 \tag{374}$$

$$\frac{d^2x}{dt^2} = \frac{-\omega^2\cos(r,\omega t)}{r^2} \tag{375}$$

$$\frac{d^2y}{dt^2} = \frac{d\left(\left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t}\frac{d(\omega t)}{dt}\right)}{dt} \tag{376}$$

$$\frac{d^2y}{dt^2} = \left(\frac{\partial^2\sin(r,s)}{\partial s^2} \cdot \left.\frac{ds}{dt}\right|_{s=\omega t}\right) \cdot \frac{d(\omega t)}{dt} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{377}$$

$$\frac{d^2y}{dt^2} = \left(\frac{-\sin(r,\omega t)}{r^2} \cdot \frac{d(\omega t)}{dt}\right) \cdot \frac{d(\omega t)}{dt} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{378}$$

$$\frac{d^2y}{dt^2} = \frac{-\sin(r,\omega t)}{r^2} \cdot \left(\frac{d(\omega t)}{dt}\right)^2 + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{379}$$

$$\frac{d^2y}{dt^2} = \frac{-\sin(r,\omega t)}{r^2} \cdot \omega^2 + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{380}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \left.\frac{\partial\sin(r,s)}{\partial s}\right|_{s=\omega t} \cdot \frac{d^2(\omega t)}{dt^2} \tag{381}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \frac{\cos(r,\omega t)}{r} \cdot \frac{d^2(\omega t)}{dt^2} \tag{382}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + \frac{\cos(r,\omega t)}{r} \cdot 0 \tag{383}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} + 0 \tag{384}$$

$$\frac{d^2y}{dt^2} = \frac{-\omega^2\sin(r,\omega t)}{r^2} \tag{385}$$

$$\vec{a}(t) = \left(\frac{-\omega^2\cos(r,\omega t)}{r^2}, \frac{-\omega^2\sin(r,\omega t)}{r^2}\right) \tag{386}$$

$$\vec{a}(t) = \frac{-\omega^2\left(\cos(r,\omega t), \sin(r,\omega t)\right)}{r^2} \tag{387}$$

$$\vec{a} = \frac{-\omega^2\vec{r}}{r^2} \tag{388}$$

$$a = |\vec{a}| \tag{389}$$

$$a = \left|\frac{-\omega^2\vec{r}}{r^2}\right| \tag{390}$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|} \tag{391}$$

$$a = \frac{|-\omega^2| \, |\vec{r}|}{|r^2|} \tag{392}$$

$$a = \frac{\omega^2 \, |\vec{r}|}{|r^2|} \tag{393}$$

$$a = \frac{\omega^2 r}{|r^2|} \tag{394}$$

$$a = \frac{\omega^2 r}{r^2} \tag{395}$$

$$a = \frac{\omega^2}{r} \tag{396}$$

$$F = ma \tag{397}$$

$$F = m\frac{\omega^2}{r} \tag{398}$$

$$F = \frac{m\omega^2}{r} \tag{399}$$

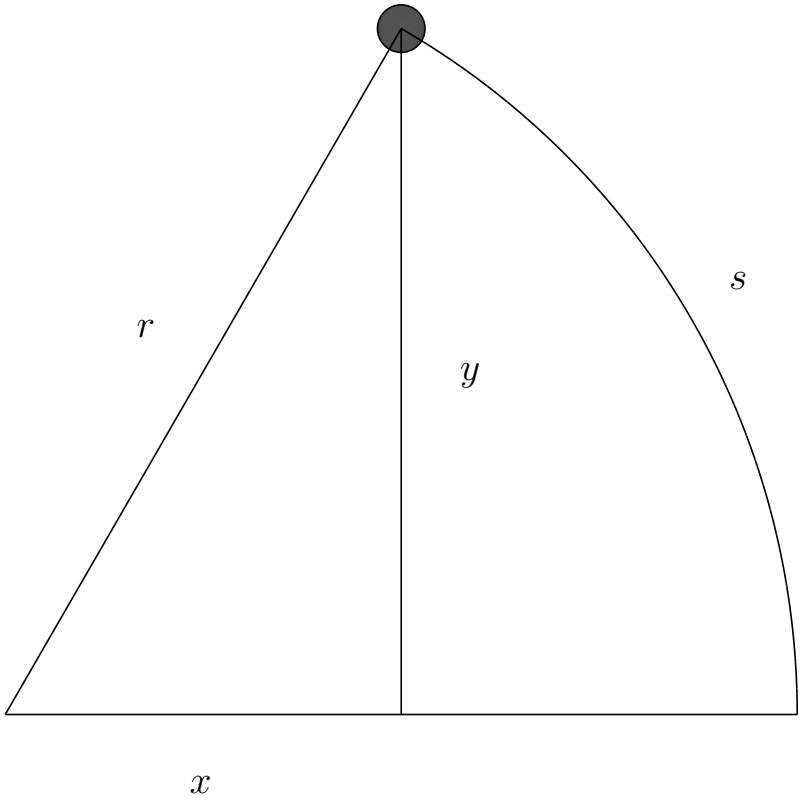


Figure 4: Uniform circular motion.

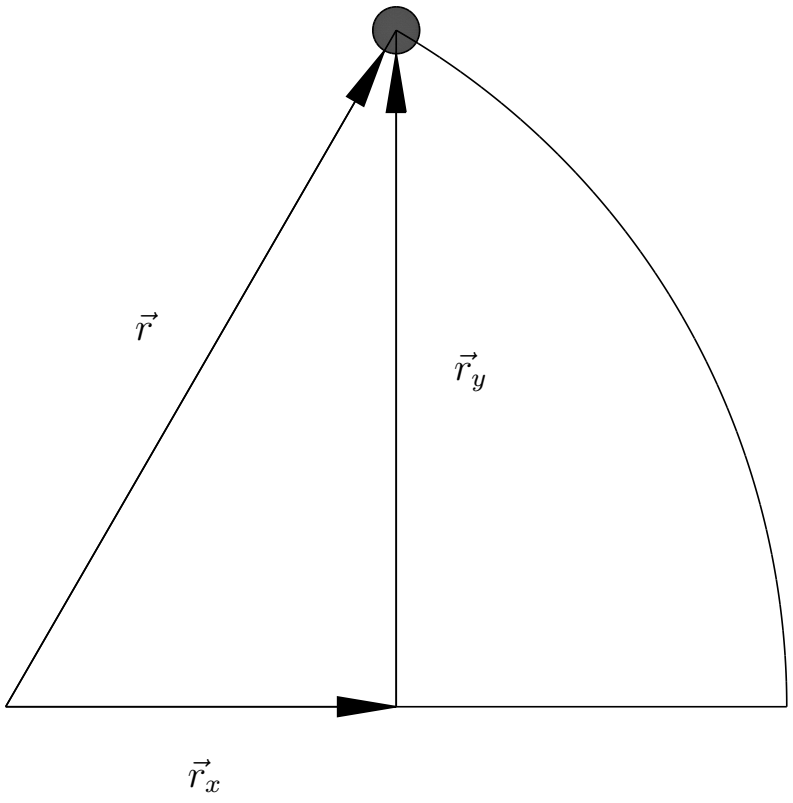


Figure 5: Uniform circular motion (vector diagram).